Solution:

Since X denotes the total score in two throws, it takes on the values 2, 3, 4, 5, and 6.

From the Sample space **S**, we have

Values of the Random Variable	2	3	4	5	6	Total
Number of elements in inverse images	1	4	10	12	9	36

Sample space S

I \ II	1	2	2	3	3	3
1	2	3	3	4	4	4
2	3	4	4	5	5	5
2	3	4	4	5	5	5
3	4	5	5	6	6	6
3	4	5	5	6	6	6
3	4	5	5	6	6	6

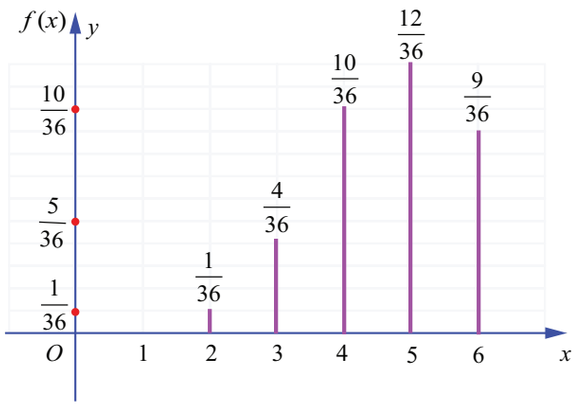
$P(X=2) = \frac{1}{36}$, $P(X=3) = \frac{4}{36}$

$P(X=4) = \frac{10}{36}$, $P(X=5) = \frac{12}{36}$, and

$P(X=6) = \frac{9}{36}$.

(i) Probability mass function is

x	2	3	4	5	6
$f(x)$	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{10}{36}$	$\frac{12}{36}$	$\frac{9}{36}$



Probability mass function

Fig. 11.8

(ii) Cumulative distribution function

By definition of the cumulative distribution function for discrete random variable we have

$$F(x) = P(X \le x) = \sum_{x_i \le x} P(X = x_i),$$

$P(X < x) = 0$ for $-\infty < X < 2$.

$$F(2) = P(X \le 2) = \sum_{-\infty}^{2} P(X = x) = P(X < 2) + P(X = 2) = 0 + \frac{1}{36} = \frac{1}{36}.$$

$$F(3) = P(X \le 3) = \sum_{-\infty}^{3} P(X = x) = P(X < 2) + P(X = 2) + P(X = 3) = 0 + \frac{1}{36} + \frac{4}{36} = \frac{5}{36}.$$

$$F(4) = P(X \le 4) = \sum_{-\infty}^{4} P(X = x) = P(X < 2) + P(X = 2) + P(X = 3) + P(X = 4)$$

$$= 0 + \frac{1}{36} + \frac{4}{36} + \frac{10}{36} = \frac{15}{36}.$$

$$F(5) = P(X \le 5) = \sum_{-\infty}^{5} P(X = x) = P(X < 2) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$$

$$= 0 + \frac{1}{36} + \frac{4}{36} + \frac{10}{36} + \frac{12}{36} = \frac{27}{36}.$$

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